SIRS Criteria Assessment

SIRS Criteria are met if 2 or more of the following are present:

- Temperature < 36 or > 38.3°C
- Heart rate > 90
- Respiratory rate > 20
- Acutely confused

If meningitis suspected ensure appropriate PPE is worn; Mask and goggles

YES

Routine Vital Sign monitoring, plus:

- ECG
- ETCO2
- SpO2

NO

Routine Vital Sign Monitoring

Fever Present?

Consider Paracetamol

- **Adults:** 1g IV/PO
- **Pediatrics:** 15 mg/kg IV/PO (max 1 g).

EMT

Signs of Poor Perfusion?

Administer Crystalloid Fluid

- **Adults: 500-1000 mL bolus IV** (repeat as needed, max 2000mL)
- **Pediatrics: 20 mL/kg bolus IV** (repeat as needed, max 60 mL/kg).

EMT

Reassess

Signs of Poor Perfusion Remain after:
2000ml (adults) or 40ml/kg (pediatrics) of fluid

Consider IV Adrenaline

AP

	ADULT	PEDIATRIC
Push Dose	0.01mg IV/IO Every 2-5minutes	0.005mg/kg IV/IO Every 2 – 5 minutes
Infusion	2-10mcg/min IV	0.05-0.3mcg/kg/min IV

Patient Care Goals

1. Identify and manage sepsis or septic shock.
2. Identify patients at risk of sepsis.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients presenting with fever or suspected sepsis, including:
  - Documented or reported fever (temperature >38°C) or hypothermia (<36°C).
  - Symptoms suggestive of infection (e.g., chills, rigors, malaise, or localized symptoms like cough, dysuria, or wound infection).
  - Signs of systemic inflammatory response or sepsis, such as tachycardia, tachypnea, hypotension, or altered mental status.
  - Suspected septic shock with hypotension (SBP <90 mmHg in adults, or age-specific for pediatrics) and signs of poor perfusion.

Exclusion Criteria

- Fever or symptoms attributable to a non-infectious cause (e.g., heat stroke, or drug reaction).

Key Patient Assessment Points

1. **Infection History**
   - Identify potential sources of infection (e.g., respiratory, urinary, skin, or abdominal).
   - Inquire about recent illness, surgery, invasive procedures, or immunocompromised states (e.g., diabetes, cancer, or HIV).
   - Confirm history of fever, chills, or rigors, including duration and severity.
2. **Sepsis Screening**
   - Assess for systemic inflammatory response syndrome (SIRS) criteria:
     - Temperature >38°C or <36°C.
     - Heart rate >90 beats/min (or age-specific for pediatrics).
     - Respiratory rate >20 breaths/min (or age-specific for pediatrics).
     - Altered mental status or confusion.
   - Evaluate for signs of organ dysfunction (e.g., decreased urine output, mottled skin, or hypoxia).
3. **Temperature Assessment**
   - Measure body temperature to confirm fever (>38°C) or hypothermia (<36°C).
   - Note subjective reports of fever if measurement is unavailable.
4. **Source Identification**
   - Look for localized signs of infection (e.g., erythema, purulent discharge, or cough with productive sputum).
   - Check for medical devices (e.g., catheters, central lines) as potential infection sources.
5. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.

Patient Management & Interventions

1. **Primary and Secondary Surveys**
   - Perform Primary and secondary survey as per Universal Care CPG.
   - **Consider Oxygen**
     - Titrate to maintain SpO2 ≥94% if hypoxia or respiratory distress is present.
2. **Consider Red Flags**
   - Hypotension (SBP <90 mmHg in adults, or age-specific for pediatrics) despite fluid resuscitation.
   - Altered mental status or unresponsiveness.
   - Severe respiratory distress or hypoxia (SpO2 <90%).
   - Signs of septic shock (e.g., mottled skin, delayed capillary refill >2 seconds).
   - Immunocompromised patients with fever and systemic symptoms.
3. **Evaluate for Signs of Shock or Instability**

- Hypotension, tachycardia, or poor perfusion (e.g., cool, clammy skin, delayed capillary refill).
- If septic shock is suspected, initiate aggressive fluid resuscitation and prioritize transport.

4. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
5. **Establish IV Access**
   - Indicated for hypotension, suspected sepsis, or anticipated fluid/medication administration.
   - Consider intraosseous (IO) access if IV access cannot be obtained.
6. **Fluid Resuscitation**
   - Administer Normal Saline or Ringers Lactate for signs of sepsis or septic shock:
     - Adults: 500-1000 mL bolus IV (up to 2000mL), repeat as needed.
     - Pediatrics: 20 mL/kg bolus IV, reassess, repeat as needed (max 60 mL/kg).
   - Monitor for fluid overload, especially in patients with cardiac or renal comorbidities.
7. **Paracetamol**
   - Indicated to reduce fever and improve patient comfort.
     - Adults: 1 g IV/PO
     - Pediatrics: 15 mg/kg IV/PO (max 1 g).
8. **Consider Vasopressors**
   - **Consider Adrenaline**
   - Indications:
     - Fluid Refractory hypotension.
   - Adrenaline doses should be titrated to a maintain a MAP of >65, the improvement of shock symptoms or the return of urine output.
     - IV Push Dose:
       - Adult: 0.01mg every 2-5 minutes, as required.
       - Pediatric: 0.005mg/kg every 2-5 minutes, as required.
     - IV Infusion:
       - Adult: 2-10mcg/min
       - Pediatric: 0.05-0.3mcg/kg/min
9. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.

Patient Safety Considerations

- Adrenaline Administration:
  - Use a syringe driver wherever possible.
  - IV adrenaline infusions through an IV administration set may be used provided that:
  - Drip rate calculations are cross checked.
  - Extreme caution is exercised to unsure the rate is not mistakenly changed.
  - In children younger than 5 years old, and when a syringe driver is not available, IV push dose adrenaline is preferred over IV adrenaline infusion via a drip set.
  - Adrenaline should be reduced in patients with underlying cardiac disease.
- Transportation Decision Making:
  - Patients who meet SIRS criteria must be recommended transfer to hospital.

Clinical Insights & Additional Guidance

- Caution should be used when assessing the elderly and pediatrics with fever as some symptoms may not present or may be masked by underlying conditions.
- In the elderly:
  - Classic symptoms such as fever and chills may present as lethargy or decreased appetite
  - Fever response may be impaired
  - Altered mental status may be masked by other diseases such as dementia
  - Tachycardia may be masked by beta-blockers

- Respiratory rates may already be elevated due to lung disease
- Baseline hypertension may mask hypotension
- Clearly identify to dispatch 2 that the patient meets SIRS criteria when coordinating emergency department admission.

Documentation

- Identification: Patient's full name, age, and gender.
- Mechanism / Medical Complaint: Symptoms such as fever, chills, altered mental status, hypotension, or signs of infection.
- Injuries / Information: History of recent surgery, device use (e.g., catheter), immunosuppression, or other infection risks.
- Signs: Vital signs (BP, HR, RR, SpO_2), temperature, mental status (AVPU/GCS), signs of poor perfusion (e.g., capillary refill, mottled skin).
- Treatment:
  - IV/IO access placement and fluid resuscitation details (type, volume, response).
  - Paracetamol: dose, route, time.
  - Oxygen therapy and target SpO_2.
  - Adrenaline if given: dose, route, rate, response.
- Allergies: All known allergies, especially to medications (e.g., paracetamol, antibiotics).
- Medications: Current or recent use of immunosuppressants or antibiotics.
- Background: Chronic illness (e.g., diabetes, cancer, renal disease), recent hospitalization.
- Other:
  - SIRS criteria and shock findings.
  - Dispatch and hospital coordination notes.
  - Document any deterioration or red flag conditions.

Key Performance Indicators

1. Assess and document SIRS criteria for all suspected sepsis patients.
2. Assess and document temperature for all suspected sepsis patients.
3. Administer fluid for hypotension (where appropriate).

References

1. National Association of State EMS Officials (NASEMSO) (2022) *National Model EMS Clinical Guidelines*. Version 3. Available at: https://nasemso.org/wp-content/uploads/National-Model-EMS-Clinical-Guidelines_2022.pdf (Accessed: 28 July 2025).ems.utah.gov
2. Joint Royal Colleges Ambulance Liaison Committee (JRCALC) (2025) *Sepsis guideline*. JRCALC Plus and iCPG apps. Available at: https://jrcalc.org.uk/ (Accessed: 28 July 2025).linkedin.com
3. BC Emergency Health Services (BCEHS) (2025) *Clinical Practice Guidelines: K02 Sepsis*. BCEHS Handbook. Available at: https://handbook.bcehs.ca/clinical-practice-guidelines/k-infectious-diseases/k02-sepsis/ (Accessed: 28 July 2025).handbook.bcehs.ca
4. St John New Zealand (2025) *Bacterial Sepsis - NZ CPG*. Clinical Procedures and Guidelines. Available at: https://cpg.stjohn.org.nz/tabs/guidelines/infection-eas/page/bacterial-sepsis-eas (Accessed: 28 July 2025).cpg.stjohn.org.nz
5. Queensland Ambulance Service (QAS) (2025) *Clinical Practice Guidelines: Medical/Sepsis*. Digital Clinical Practice Manual. Available at: https://www.ambulance.qld.gov.au/__data/assets/pdf_file/0019/219052/cpg_sepsis.pdf (Accessed: 28 July 2025).ambulance.qld.gov.au
6. Evans, L., Rhodes, A., Alhazzani, W., Antonelli, M., Coopersmith, C.M., French, C., Machado, F.R., McIntyre, L., Ostermann, M., Prescott, H.C., Schorr, C., Simpson, S., Joost Wiersinga, W., Alshamsi, F., Angus, D.C., Arabi, Y., Azevedo, L., Beale, R., Beilman, G., Belley-Cote, E., Burry, L., Cecconi, M., Centofanti, J., Yataco, A.C., De Waele, J., Dasta, J.F., Du, B., Dubler, S., Ferrer, R., Gomersall, C.D., Hodgson, C., Hylander Møller, M., Iwashyna, T., Jacob,

S., Kleinpell, R., Klompas, M., Koh, Y., Kumar, A., Kwizera, A., Lobo, S., Masur, H., McGloughlin, S., Mehta, S., Mehta, Y., Mer, M., Nunnally, M., Oczkowski, S., Osborn, T., Papathanassoglou, E., Perner, A., Puskarich, M., Roberts, J., Schweickert, W., Seckel, M., Sevransky, J., Sprung, C.L., Welte, T., Zimmerman, J. and Levy, M. (2021) *Surviving sepsis campaign: international guidelines for management of sepsis and septic shock 2021*. *Intensive Care Medicine*, 47(11), pp.1181-1247.pmc.ncbi.nlm.nih.gov

7. Green, R.S., Travers, A.H., Cain, E., Campbell, S.G., Jensen, J.L., Petrie, D.A., Erdogan, M., Patrick, G. and Patrick, W. (2008) *Canadian Association of Emergency Physicians Sepsis Guidelines: the optimal management of severe sepsis in Canadian emergency departments*. *CJEM*, 10(5), pp.443-459.pmc.ncbi.nlm.nih.gov
8. Polito, C.C. and Martin, G.S. (2024) *Septic shock in the prehospital setting: a scoping review*. *Scandinavian Journal of Trauma, Resuscitation and Emergency Medicine*, 32(1), p.101.sjtrem.biomedcentral.com
9. Kleinman, M.E., Goldberger, Z.D., Rea, T., Swor, R.A., Bobrow, B.J., Brennan, E.E., Terry, M. and Hemphill, J.C. (2025) *Evaluation and management of suspected sepsis and septic shock in adults*. UpToDate. Available at: https://www.uptodate.com/contents/evaluation-and-management-of-suspected-sepsis-and-septic-shock-in-adults (Accessed: 28 July 2025).uptodate.com
10. Prescott, H.C. and Angus, D.C. (2020) *Sepsis: Diagnosis and Management*. *American Family Physician*, 101(7), pp.409-418.aafp.org
11. Alam, N., Oskam, E., Stassen, P.M., Exter, P.V., van de Ven, P.M., Haak, H.R., Holleman, F., van Zanten, A., van Beek, E.J.R., Manten, W., Kramer, M.H.H., Nanayakkara, P.W.B. and PHANTASi Trial Investigators (2018) *Prehospital antibiotics in the ambulance for sepsis: a multicentre, open label, randomised trial*. *The Lancet Respiratory Medicine*, 6(1), pp.40-50.sciencedirect.com
12. Groenewoudt, M., Roest, A.A., Leijten, F.M. and Stassen, P.M. (2019) *Prehospital characteristics among patients with sepsis: a prospective cohort study*. *BMC Emergency Medicine*, 19(1), p.43.bmcemergmed.biomedcentral.com
13. Rhodes, A., Evans, L.E., Alhazzani, W., Levy, M.M., Antonelli, M., Ferrer, R., et al. (2017) *Surviving Sepsis Campaign: International Guidelines for Management of Sepsis and Septic Shock: 2016*. *Intensive Care Medicine*, 43(3), pp.304-377.link.springer.com
14. Cannon, C.M., Holthaus, C.V., Zubrow, M.T., Posa, P., Gunaga, S., Kella, V., Elkin, R., Davis, S., Turman, B., Weingarten, J., Milling, T.J. and Lidsky, N. (2018) *The GENESIS project (GENeralized Early Sepsis Intervention Strategies): a multicenter quality improvement collaborative*. *Journal of Intensive Care Medicine*, 33(12), pp.699-707.jems.com
15. Jouffroy, R., Gueye, P., Saint Martin, L.C., Jacquin, L., Vaittinada Ayar, P., Khoury, A., Lemoine, F., Bloch-Laine, E., Bourgoin, C., Dubost, C., Aghasaryan, A., Khalil, A., Tazarourte, K. and Vivien, B. (2023) *Prehospital administration of antibiotics in addition to usual care versus usual care alone for sepsis patients: a retrospective cohort study*. *Therapeutic Advances in Infectious Disease*, 10